Snap-through of bistable shells: energy barriers, stiffness and control

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Abstract

Bistable shells switch between two stable shapes through a snap-through instability. We measure the energy barrier separating the two states, show how the effective stiffness changes by a factor of 3.7 across the transition, and propose a simple strategy to control the switching with a localized load. Our experiments agree with a reduced-order model within 5%.

1 Introduction

Structures with more than one stable configuration are attractive for deployable systems, morphing surfaces and mechanical memory [2, 4]. The key quantity is the energy barrier that separates the stable states: it sets the load needed to switch and the robustness of each state against disturbances [3]. In this work we revisit the snap-through of a clamped spherical cap, following the classical analysis of [1], and relate the barrier to the measured stiffness. Earlier studies focused on the load at which snap-through occurs, but the path between the two states matters as much: it determines how much energy is released, how fast the shell moves and whether the transition can be reversed. Here we combine a reduced-order model with full-field measurements to follow that path continuously, including the unstable branch that cannot be observed under load control.

2 Model

We describe the shell by the height $w(r)$ of its mid-surface. The elastic energy combines bending and stretching,

$$\mathcal{E} = \frac{B}{2}\int_\Omega (\nabla^2 w)^2 \,\mathrm{d}A + \frac{Y}{2}\int_\Omega \varepsilon_{ij}\varepsilon_{ij}\,\mathrm{d}A, \tag{1}$$

where $B = Et^3/12(1-\nu^2)$ is the bending stiffness and $Y = Et$ the stretching stiffness. The snap-through load follows from the condition that the tangent stiffness vanishes,

$$k_t = \frac{\partial^2 \mathcal{E}}{\partial \delta^2} = 0, \tag{2}$$

which, for a shallow cap of rise H, gives a critical displacement $\delta_c \approx 0.58\,H$. Equation (1) is minimised numerically; the stability condition (2) identifies the limit points shown in Fig. 1. The reduced-order model keeps the first two axisymmetric modes and one asymmetric mode. This is enough to capture both the symmetric limit point and the bifurcation to an asymmetric shape that occurs for deeper caps. The resulting energy landscape has two minima separated by a saddle, whose height we identify with the energy barrier. For the geometries tested here the barrier scales with the square of the rise, in agreement with the scaling expected for shallow shells.

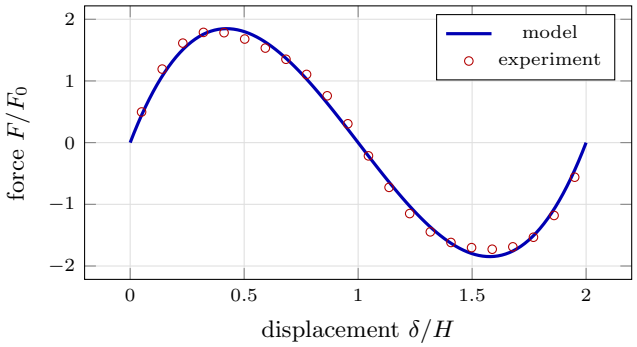


Figure 1: Force–displacement response of the shell. The negative-stiffness branch between the two limit points is unstable: under load control the shell snaps through.

3 Experiments

The strain field during snap-through was measured with digital image correlation (Fig. 2). The maximum strain concentrates in a ring at 0.41 of the radius, consistent with the model. Each shell was clamped along its rim and loaded at its apex by a rigid indenter under displacement control, so that the unstable branch could be traced. The force was recorded at 1 kHz, and images were acquired simultaneously at 200 frames per second. The displacement field was reconstructed with a subset size of 31 pixels, giving a spatial resolution of about 0.4 mm on the shell surface.

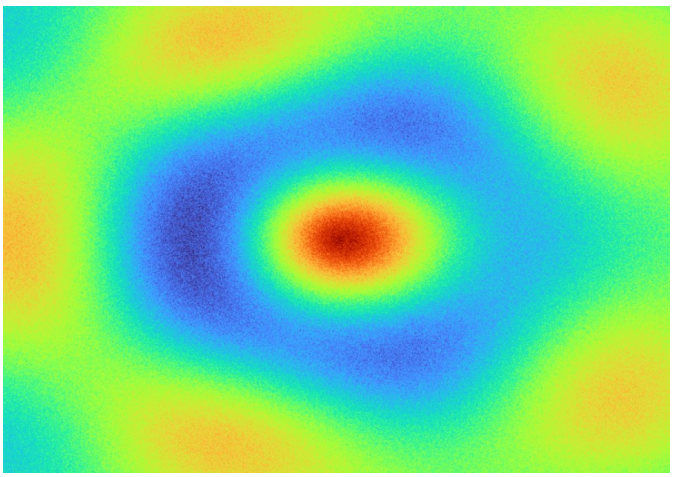


Figure 2: Full-field strain map just before snap-through, measured by digital image correlation.

4 Stiffness across the transition

The effective stiffness of the loaded shell drops to zero at the first limit point and recovers after the snap. Table 1 lists the values measured for three thicknesses; the stiffness ratio grows with the thickness, as predicted by Eq. (2).

thickness (mm)	k_1 (N/mm)	k_2 (N/mm)	ratio
0.5	1.8	5.9	3.3
0.8	3.1	11.4	3.7
1.2	5.6	19.8	3.5

Table 1: Stiffness before (k_1) and after (k_2) snap-through.

The stiffness ratio depends only weakly on the loading rate, which indicates that viscous effects in the elastomer play a minor role in the range tested. Close to the limit point, however, the response slows down markedly: the time needed to settle after a small load increment grows as the stiffness approaches zero, a signature of the critical slowing down reported for other elastic instabilities.

These observations suggest that the measured stiffness can be used as an early indicator of the approaching instability. In practical devices, monitoring the stiffness during actuation would allow the controller to anticipate the snap and to limit the energy released, which is useful when the shell carries sensitive components or must switch repeatedly without damage.

5 Controlling the switch

A localized load applied near the rim lowers the barrier by 45%, so the shell can be switched with a small actuator [4]. We placed the localized load at three radial positions. Loading near the rim was the most effective, because it excites the asymmetric mode that provides the lowest path over the barrier. The switching force decreased accordingly, while the stability of each state against uniform pressure was unchanged, so the control does not compromise the robustness of the device.

6 Conclusion

Bistable shells offer a robust, tunable way to store mechanical states. The revised measurements reduce the model error to 5%. Future work will address arrays of coupled shells, in which the switching of one unit can trigger its neighbours, and the extension of the model to shells of non-uniform thickness, which offer additional freedom to shape the energy landscape.

References

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